



Comparison between intrauterine application of an antibiotic and an herbal product to treat clinical endometritis in dairy cattle – A randomized multicentre field study

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ABSTRACT

Uterine diseases are main indications for antibiotic use in dairy cows. To test a non-antibiotic treatment option, we compared the effect of an intrauterine cephalosporin (Metricure®; cefapirin benzathin 500 mg per dosis; CEPH) with an intrauterine applied herbal product (25 ml of EucaComp® PlantaVet containing alcoholic extracts of *Calendula officinalis* L., *Melissa officinalis* L., *Origanum majorana* L. and *Eucalyptus globulus* Labill. (EUC)) on the clinical cure of endometritis.

Examinations of 816 cows between 21 and 35 days after calving were performed and cases of clinical endometritis ($n = 169$) were included. Diagnosis based on a scoring system for vaginal discharge. Study animals were randomly assigned to one of two treatment groups and treated immediately. After excluding animals with incomplete datasets, 136 cows (EUC: $n = 61$; CEPH: $n = 75$) remained for the final analysis.

In total, 64% (EUC: 61%, CEPH: 67%) of analysed endometritis cases were considered as clinically cured 14 ± 2 days after the first treatment, 15% stayed uncured after the application of a maximum of two consecutive treatments, leading to an overall clinical cure rate of 85% (EUC: 82%, CEPH: 88%). No statistically difference in clinical cure rates could be observed between both treatment groups nor 14 ± 2 days after the first treatment ($p = 0.956$) neither regarding the overall cure rate ($p = 0.923$).

In conclusion, the clinical cure of dairy cows' endometritis after the intrauterine application of the herbal product was non-inferior to the intrauterine application of the antibiotic cephalosporin. These results could contribute to reduce the antimicrobial use in the daily veterinary routine treatment of endometritis.

1. Introduction

Reproductive disorders in dairy cattle represent a challenging problem for farmers all around the world due to their strong detrimental effect on fertility and their economic impact (LeBlanc et al., 2002a). Clinical endometritis is defined as the presence of purulent uterine discharge detectable in the vagina without any signs of systemic illness in cows 21 days or more postpartum (Sheldon et al., 2009). Routine therapy of clinical endometritis is based on the parenteral administration of prostaglandins or intrauterine administration of antibiotics like

beta-lactams or oxytetracyclines (Galvão et al., 2009; Denis-Robichaud and Dubuc, 2015; Hehenberger et al., 2015), however, a “gold standard therapy” is still lacking (Dhaliwal et al., 2001; Brodzki et al., 2018; Brodzki et al., 2020).

Emergence of antibiotic resistances in veterinary as well as in human medicine reinforced the urgent need to develop alternative therapeutic strategies for reproductive disorders (Callaway et al., 2021; Kasimanickam et al., 2021; Casseri et al., 2022). The European Union as well as countries like Germany or Switzerland developed strategies on antibiotic resistance (Switzerland STAR; Germany DART). In the framework

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of the concept *One Health*, these strategies promote a safety use of antimicrobials and define objectives for human medicine, veterinary medicine, agriculture and the environment (StAR, 2015; DART, 2020; EU, 2019).

Phytotherapy represents an area with great potential in the context of endometritis treatment (Paiano and Baruselli, 2022; Gopikrishnan et al., 2022; Boudelal et al., 2022) and especially farmers following organic agriculture are willing to use it for this indication (Mayer et al., 2014; Pinedo and Velez, 2019). Medications based on plants have already been historically (Von Grebner Joseph, 1883; Wetterwald, 1878; Steiner et al., 2022) and are still recently used by European farmers and veterinarians (Mayer et al., 2014; Tamminen et al., 2018; Stucki et al., 2019). It is also to note that - besides plant-based products - other non-antibiotic intrauterine therapies to treat endometritis in dairy cattle have also been studied and already showed encouraging results like for example the application of ozone (Đuričić et al., 2014; Samardžija et al., 2017) or zinc oxide (Amin et al., 2023).

Intrauterine antibiotic treatment was the standard therapy of clinical endometritis used by nearly 70% of the Swiss veterinarians, most frequently with Metricure® (CEPH, cephapirin benzathin 500 mg, MSD Animal Health, Luzern, Switzerland; (Hehenberger et al., 2015)). In many other countries cephapirin is widely used for the treatment of clinical endometritis (Tison et al., 2017; Barański et al., 2022; Denis-Robichaud and Dubuc, 2015; Madoz et al., 2017). The first-generation antibiotic cephapirin has a broad-spectrum antimicrobial action against sensitive gram-positive and gram-negative anaerobic bacteria. Its successful use in the intrauterine treatment of clinical endometritis has been shown in several studies (Dohmen et al., 1995; Runciman et al., 2009; Denis-Robichaud and Dubuc, 2015; Tison et al., 2017; Appendix 1) although not in all (LeBlanc et al., 2002b; Ahmadi and Dehghan, 2007).

EucaComp®PlantaVet (EUC, SaluVet, Bad Waldsee, Germany, 250 ml: *Calendula officinalis* L., *Melissa officinalis* L., *Origanum majorana* L., *Eucalyptus Globulus* Labill.) is a pure herbal veterinary medicinal product registered for the intrauterine treatment of endometritis and on the market since many decades in Germany. Plants contained in this drug present together antimicrobial, anti-inflammatory, antioxidative, immunostimulatory and wound-healing properties and are typically reflecting the multitarget aspect of herbal products (Kern et al., 2013). Only few studies exist about this commercial product and clinical studies comparing EUC and CEPH directly are recently not available (Appendix 2). However, some studies already showed positive effects of EUC on reproductive performance in comparison to untreated cows (Schnellbach, 1991; Spielberger, 1991), to cows treated with intrauterine Lotagen®Solution (MSD Santé Animale, Beaucouze Cedex, France, 100 ml) (Heuwieser et al., 2000; Handler et al., 2005; Kaya, 2008) or oxytetracycline (Bogdan et al., 2009).

The aim of this study was to provide for the first time a clinical study to directly compare under field conditions the two registered veterinary medicinal products for the intrauterine treatment of endometritis EUC and CEPH regarding their effect on endometritis' clinical cure (clear vaginal mucus or, in the absence of vaginal discharge, normal uterine size without any ultrasound-assessed pathological uterine content). Our hypothesis was that the phytotherapeutic preparation EUC leads to non-inferior clinical cure rates compared with CEPH, at least after two consecutive treatments at an interval of two weeks.

2. Material & methods

A multicentre randomized controlled clinical study has been conducted under typical veterinary practice field conditions characterised by smallholder dairy farms. The efficacy of an antibiotic intrauterine treatment and an herbal veterinary medical intrauterine product has been compared. This prospective study was performed with permission and in accordance with Swiss law. The following approval number was allocated by the animal experimentation commission (elected by the

cantonal executive council): BE 9/20 +.

2.1. Animals, farms, housing

A total of 31 farms were included in this project. All of them were located in Switzerland and were taking part in the herd management program provided by veterinarians of the "Clinique du Vieux Château". A flyer was distributed to all farms and those handing in an informed consent were included.

Involved farms were characterised by different dairy management types. Herd sizes ranged from 20 to 120 cows. Three farms were managed according to the Swiss organic regulation (BioSuisse, 2021), whereas the others were non-organic farms. Cows of the breeds Holstein, Swiss Fleckvieh, Brown Swiss, Montbéliarde and Jersey were included. In 16 farms, cows were kept in tie-stall housing, whereas the others had loose housing barns. From spring to autumn, cows of all herds had access to pasture. Cows were milked twice a day in all the farms, except on one farm, which used a milking robot. The average yearly milk yield of the herds varied from 6000 kg to 11,000 kg per cow and year.

2.2. Inclusion / exclusion criteria

During the entire study period (September 2019 to February 2022), a total of 816 examinations of cows were conducted between 21 and 35 days after calving. Cows examined in the corresponding period and suffering from clinical endometritis grade 1–3 (Sheldon et al., 2006; detailed information see chapter 2.4.1) were included in the study.

Exclusion criteria were uterine prolapse, fetotomy, caesarean section (preceding calving), intrauterine cephapirin infusion or intramuscular prostaglandin injection before the first postpartum examination. Furthermore, cows showing fever $\geq 39.7^{\circ}\text{C}$ (Kristula et al., 2001), clinical mastitis with fever or displacement of the abomasum on the day of the examination led to an exclusion. In cases where cows were examined after two consecutive calvings within the course of the project only the first endometritis case was included in the analysis.

2.3. Anamnestic parameters

Several anamnestic parameters were collected (Table 1) including lactation number, breed, calving season, calving assistance (no assistance / light extraction help by farmer / veterinary assistance), twins (yes / no), already applied treatments between calving and first postpartum control (intrauterine applications of iodine capsules / antibiotics therapy with uterine indication / antibiotics therapy with another indication than uterus / intramuscular application of long-acting oxytocin) or postpartum complications such as retained fetal membranes (no / yes with fever / yes without fever). All treatment data were based on the mandatory recording system for veterinary treatments in Switzerland. More than one third of all cases included in the statistical analysis were in the second or third lactation ($n = 52$, 38%; Table 1; Appendix 3). High yielding dairy breeds represented about 80% of the included cases (Holstein: $n = 89$ (CEPH: 59%, EUC: 74%); Brown Swiss: $n = 19$ (CEPH: 20%, EUC: 7%); Jersey: $n = 2$ (CEPH: 1%, EUC: 1%) dual purpose breeds the remaining part (Montbéliarde: $n = 22$ (CEPH: 16%, EUC: 16%); Swiss Fleckvieh: $n = 4$ (CEPH: 4%, EUC: 2%)).

Autumn (September, October, November) was the season during which the highest proportion of cases calved ($n = 60$, 44%). Most of the calvings were without any difficulty ($n = 120$, 88%). Twin births ($n = 15$) represented 11% of the cases. After 17% of the calvings ($n = 23$), cows had suffered from placental retention (Table 1).

2.4. Study design

A total of 14 veterinarians working for the clinic were involved in the data collection. Before the start of the project, two online meetings were organised for instructions for homogeneity of diagnoses, treatments and

Table 1

Distribution of endometritis cases considered for statistical analysis between both treatment groups according to disease severity at day 21–35 postpartum and anamnestic parameters.

Treatment group	EUC	CEPH	Total
Endometritis severity*			
Grade 1	60.7% (n = 37)	49.3% (n = 37)	54.4% (n = 74)
Grade 2	26.2% (n = 16)	44.0% (n = 33)	36.0% (n = 49)
Grade 3	13.1% (n = 8)	6.7% (n = 5)	9.6% (n = 13)
Retained placenta			
	19.7% (n = 12)	14.7% (n = 11)	16.9% (n = 23)
Without fever	6.6% (n = 4)	12.0% (n = 9)	9.6% (n = 13)
With fever	13.1% (n = 8)	2.7% (n = 2)	7.4% (n = 10)
Lactation number			
First lactation (primiparous)	27.9% (n = 17)	32.0% (n = 24)	30.1% (n = 41)
Second or third lactation	44.3% (n = 27)	33.3% (n = 25)	38.2% (n = 52)
Fourth or fifth lactation	16.4% (n = 10)	28.0% (n = 21)	22.8% (n = 31)
Higher lactation number	11.5% (n = 7)	6.7% (n = 5)	8.8% (n = 12)
Intrauterine application of iodine capsules**			
	11.5% (n = 7)	13.3% (n = 10)	12.5% (n = 17)
Antibiotic therapy for uterine indication**			
	14.8% (n = 9)	8.0% (n = 6)	11.0% (n = 15)
Systemic antibiotic therapy for another indication than uterus**			
	4.9% (n = 3)	4.0% (n = 3)	4.4% (n = 6)
Intramuscular application of long-acting oxytocin**			
	19.7% (n = 12)	8.0% (n = 6)	13.2% (n = 18)
Calving season***			
Winter	13.1% (n = 8)	24.0% (n = 18)	19.1% (n = 26)
Spring	9.8% (n = 6)	16.0% (n = 12)	13.2% (n = 18)
Summer	22.95% (n = 14)	24.0% (n = 18)	23.5% (n = 32)
Autumn	54.1% (n = 33)	36.0% (n = 27)	44.1% (n = 60)
Calving assistance			
No assistance needed	95.1% (n = 58)	82.7% (n = 62)	88.2% (n = 120)
Farmer assistance needed	0% (n = 0)	10.7% (n = 8)	5.9% (n = 8)
Vet assistance needed	4.9% (n = 3)	6.7% (n = 5)	5.9% (n = 8)
Twin birth			
	11.5% (n = 7)	10.7% (n = 8)	11.0% (n = 15)
Total	44.9% (n = 61)	55.1% (n = 75)	n = 136

* Endometritis classification of the initial diagnosis referring to Sheldon et al., 2006.

** Treatments applied between calving and first postpartum control.

*** Winter: December, January, February; Spring: March, April, Mai; Summer: June, July, August; Autumn: September, October, November.

procedures. By means of pictures of different vaginal discharges, the endometritis' classification system with three severity grades was presented.

2.4.1. Clinical and gynaecological examination of the cows

During the first control 21–35 days postpartum, all cows were

examined by rectal and vaginal palpation with a gloved hand. Before each vaginal examination, the vulvar area was carefully cleaned using a 5% diluted Betadine® (povidone-iodine 10%, Mundipharma Medical Company, Hamilton/Bermuda, Branch, Succursale Basel) solution. The presence of clinical endometritis was assessed optically, based on the scoring system for vaginal discharge of Sheldon et al. (2006): 0 = clear or translucent mucus; 1 = mucus containing flecks of white or off-white pus; 2 = exudate containing <50% white or off-white mucopurulent material; and 3 = exudate containing ≥50% purulent material, usually white or yellow, but occasionally sanguineous. Cases suffering from endometritis grade 1–3 were included and an ultrasound-guided rectal examination was performed (Easy Scan, IMV Imaging Technology, L'Aigle, France).

All cows enrolled were treated immediately after the control and re-examined (including ultrasound guided rectal examination) alike 14 ± 2 days later. If they still showed signs of endometritis (grade 1–3), they received the same treatment as given before. Cows still suffering from endometritis (grade 1–3) within the third control (28 ± 4 days after the first treatment; including ultrasound guided rectal examination) were assessed as finally uncured and an alternative therapeutic option was chosen by the responsible veterinarian and recorded. Cows having developed a pyometra (accumulation of purulent or mucopurulent material within the uterine lumen and distension of the uterus in the presence of a corpus luteum and a closed cervix (Sheldon et al., 2006), diagnosed by ultrasound-guided rectal examination in absence of purulent vaginal discharge) after the first or second treatment were also assessed as finally uncured and were treated with an intramuscular injection of 5 ml prostaglandin (ENZAPROST® ad us. vet., Injektionslösung für Rinder und Schweine, Biokema SA, Crissier).

Clinical cure was defined as clear vaginal mucus (grade 0) or, in the absence of vaginal discharge, as a normal uterine size without any ultrasound-assessed pathological uterine content (Barański et al., 2022). Rectal body temperature was measured in all cases at each examination.

2.4.2. Treatment

Cows were randomly assigned to treatment groups based on the last number of the eartag. Cows with an even eartag number (0, 2, 4, 6, 8) received 25 ml of the non-diluted herbal product EUC (dosage according to manufacturer's instructions, the use of the undiluted solution for reasons of practicability). One bottle of EUC was only used once for one date in one herd and the rest of the bottle was discarded. EUC consisted of a combination of alcoholic extracts (AE) of *Calendula officinalis* L. (marigold flower; calendulae flos, 10 g AE/100 g EUC), *Melissa officinalis* L. (lemon balm; melissae herba, 10 g AE/100 g EUC) and *Origanum majorana* L. (sweet marjoram; origanum herba, 12 g AE/100 g EUC) as well as of essential oil of *Eucalyptus globulus* Labill. (Tasmanian blue gum; eucalyptus herba, 0,5 g essential oil/100 g EUC).

The second group (odd eartag number: 1, 3, 5, 7, 9) received 19 g (one injector) of the antibiotic product CEPH (cefapirin benzathin 500 mg, Macrogol (20) Cetyl ether (Cetomacrogol 1000), Macrogol ketyl stearyl ether (Eumulgin B1), hydrogenated castor oil and medium-chain triglycerides (Mygliol 812 N)), a first generation cephalosporin.

Each treatment product was administered by a sterile 54 cm-metallic uterine catheter (ProVet; Cervix-Katheter "Prof. Dr. Götze", 4 mm diameter, Luer-Konus, chrome-plated, 44 cm). Each catheter was only used for one application and then re-sterilised. The sterilisation process included firstly a bath in a disinfection solution (Sekusept Classic® powder, Ecolab Deutschland; 30 g in 1 l water) during 15 min. After rinsing under water and drying on a clean surface, each catheter was disinfected with an alcohol swab before being carefully packed in a paper covering and stored in a dry place until the next usage.

2.4.3. Data management and statistical analyses

The collected health data were compiled in a database with a health record management software (DSHR Inc., St-Hyacinthe, Quebec, Canada). A harmonised encoding system for the collected data in the

health management software was set up.

All statistical analyses were conducted using the software R (version 4.2.0; R Core Team, 2022). Sample size and power were calculated using the package “epiR” (Stevenson et al., 2022). Assuming an overall cure rate of 80% for EUC and 85% for CEPH (approximated based on the results of Handler et al. (2005) and Tison et al. (2017)), the a priori sample size was aimed to be 158 in total in order to reach a power of 80% to detect a non-inferiority with a difference of maximum 20% in overall clinical cure rate after a maximum of two consecutive treatments (alpha level = 0.05).

Logistic regression models were applied to analyse the outcome variables (Heinze and Schemper, 2002). The two outcome variables of this trial were the clinical cure two weeks after the first treatment (yes / no) and the overall clinical cure after a maximum of two treatments (yes / no). Fixed effects in the models were: Treatment group (EUC or CEPH), endometritis severity score at first postpartum control (1–3), lactation number (1–10), calving season (winter: dec, jan, feb; spring: mar, apr, may; summer: jun, jul, aug; fall: sep, oct, nov), retained fetal membranes (no / yes without fever / yes with fever), twins (no / yes), calving process and other therapies. Concerning other therapies, it was distinguished between intrauterine application of iodine capsules, antibiotic therapies (others than those with cephalosporins) for uterine pathologies, systemic antibiotic therapies for other indications and intramuscular injection of long-acting oxytocin (for each of these variables the levels were yes / no). *P*-values of the single fixed effects were calculated with

the likelihood ratio approach. The likelihood ratio method compares the full model including all fixed effects to a reduced model without the effect of interest.

3. Results

3.1. Endometritis cases and distribution of severity grades

A total of 175 clinical endometritis cases were diagnosed during 816 postpartum examinations, which represents a prevalence of 21.4%. Before the first postpartum examination 6 cases were excluded due to treatment with prostaglandin or intrauterine cephalosporins. In total, 169 clinical endometritis cases were treated and enrolled in the trial. Overall 136 (61 EUC and 75 CEPH) were finally maintained for the data analysis as *n* = 33 cases had to be excluded from the study sample for different reasons (Fig. 1; Supplement 3): in 9 cases (5 CEPH; 4 EUC) the treatment was not in accordance with the ear mark number and in other 24 cases (5 CEPH; 19 EUC) the protocol until end of observation period was incomplete or false: 7 cases were artificially inseminated before the control following first treatment occurred (2 CEPH, 5 EUC), 6 cases were artificially inseminated before the control following the second treatment occurred (0 CEPH, 6 EUC), an interruption of the control protocol was observed in 7 cases because of a missed control two weeks following treatment (1 CEPH, 6 EUC), 1 case received accidentally the wrong therapy by the second treatment (1 EUC), 3 cases were slaughtered due

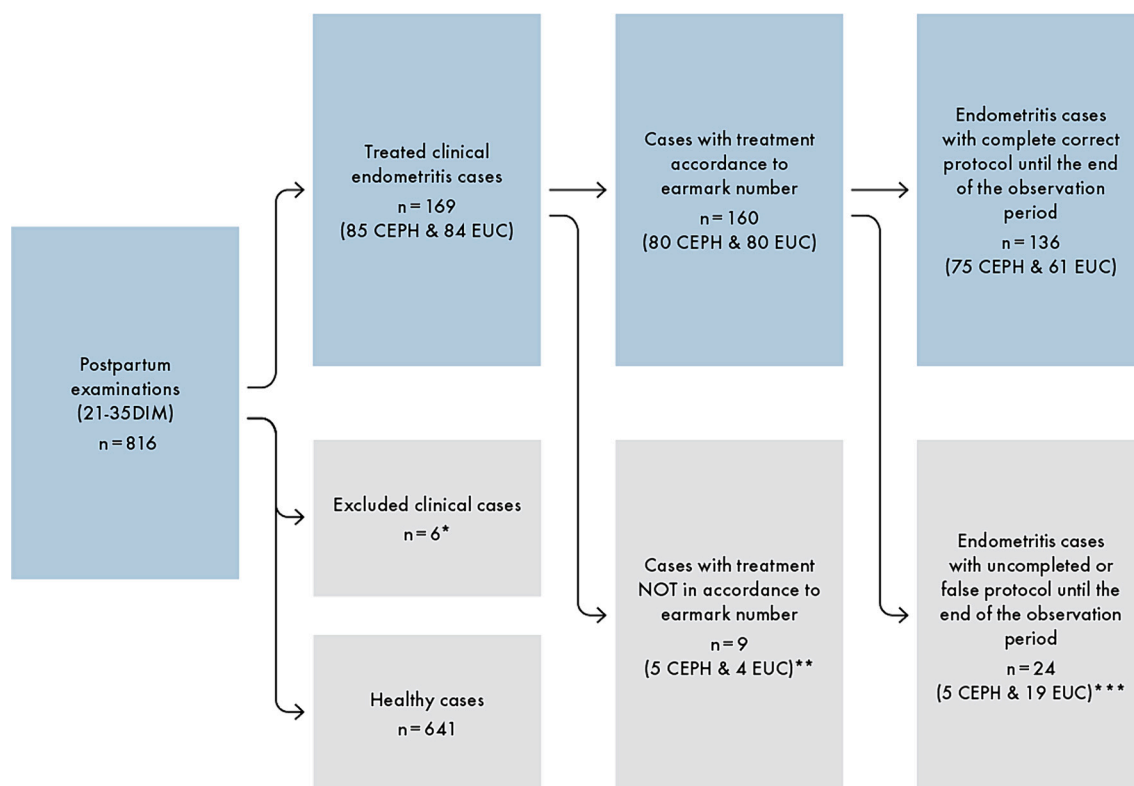


Fig. 1. Flowchart showing all examined cows, the randomization of cows with endometritis and reasons for excluding cases from analyses. DIM = days in milk, CEPH = cephalosporin antibiotic, EUC = herbal extract.

* Excluded cases concerned 5 cows having already been treated with cephalosporins before the first postpartum examination and one was treated with prostaglandin in this same interval.

** Concerning cows treated with CEPH instead of EUC, 3 were suffering from endometritis grade 1, one from endometritis grade 2 and one from endometritis grade 3. Three cows were cured after a maximum of two consecutive treatments and two were considered as resistant to therapy. Overall cure rate = 60%.

Concerning cows treated with EUC instead of CEPH, 3 of them suffered from endometritis grade 2 and one from endometritis grade 3. A cure was observed in 3 cows whereas one was considered as resistant to therapy after a maximum of two consecutive treatments. Overall cure rate = 75%.

*** It represents 24 cases that were excluded due to an interrupted or false protocol: 7 cases were artificially inseminated before the control following first treatment occurred (2 CEPH, 5 EUC), 6 cases were artificially inseminated before the control following the second treatment occurred (0 CEPH, 6 EUC), an interruption of the control protocol was observed in 7 cases because of a missed control two weeks following treatment (1 CEPH, 6 EUC), 1 case received accidentally the wrong therapy by the second treatment (1 EUC), 3 cases were slaughtered due to other reasons (2 CEPH, 1 EUC).

to other reasons (2 CEPH, 1 EUC).

Table 1 shows that 54.4% of the 136 analysed cows were suffering from an endometritis' grade 1. Severity grade 3 was observed in 9.6% of the cases.

3.2. Clinical cure

3.2.1. Clinical cure rate and clinical cure probability 14 (±2) days after first treatment

In total, 64% (n = 87) of all treated endometritis cases were considered as clinically cured two weeks after the first treatment. This proportion is comparable in both treatment groups (EUC: 61%, CEPH: 67%; Fig. 2). No statistical difference between both treatment groups (EUC and CEPH) was detected regarding clinical cure probability two weeks after the first treatment (Table 2; χ^2 (1) = 0.003; p = 0.956). Considering the actual sample size and the detected difference between treatments, the power to show non-inferiority with a maximum difference of 20% was 52%.

A pyometra was observed in four cows two weeks after the first treatment (EUC: 2 cases, CEPH: 2 cases). Endometritis severity at the starting point did not significantly affect the healing probability two weeks after the first treatment (p = 0.793). However, three other fixed effects were found to have an influence on the healing probability two weeks after the first treatment: retained placenta (χ^2 (2) = 6.713; p = 0.035), calving aid (χ^2 (2) = 6.089; p = 0.048) and intrauterine application of iodine capsules between calving and diagnosis of endometritis (χ^2 (1) = 5.652; p = 0.017). Placental retention had a negative impact on cure rate by reducing the clinical cure probability two weeks after the first treatment (Table 2; odds ratio of placental retention without fever: 0.14, odds ratio of placental retention with fever: 0.08). Totally, 4/13 cases having suffered from retained fetal membranes without fever and 3/10 of those having suffered from retained fetal membranes with fever cured after one treatment. Light calving assistance by the farmer negatively influenced the clinical cure probability two weeks after treatment (odds ratio 0.59). On the contrary, veterinary calving assistance had a positive effect on the clinical cure probability two weeks after the first treatment (odds ratio 10.7). The number of cases having needed

Table 2
Odds ratio (OR), 95% confidence intervals (CI) and p-values for the clinical cure rate 14 (±2) days after first treatment regarding treatment, pre-treatment endometritis severity and anamnestic parameters.

	OR	Lower 95% CI	Upper 95% CI	p-value
Intercept	3.913	1.135	15.327	
Treatment CEPH	1.023	0.448	2.325	0.956
Endometritis severity score 2	0.929	0.392	2.253	
Endometritis severity score 3	0.626	0.159	2.462	0.793
Retained fetal membranes without fever	0.144	0.020	0.706	
Retained fetal membranes with fever	0.081	0.002	1.444	0.035
Lactation number	0.950	0.776	1.164	0.613
Calving season spring	0.353	0.090	1.303	
Calving season summer	0.846	0.243	2.836	0.431
Calving season fall	0.627	0.204	1.781	
Calving assistance (assistance by farmer)	0.592	0.112	3.861	
Calving assistance (veterinary assistance)	10.700	1.444	271.113	0.048
Twins	0.490	0.120	1.872	0.294
Intrauterine application iodine capsules*	5.670	1.327	35.073	0.017
Antibiotic therapy for uterine indication*	2.543	0.245	49.350	0.454
Systemic antibiotics therapy for another indication than uterus*	0.749	0.121	5.768	0.762
Intramuscular application of long-acting oxytocin*	0.334	0.076	1.287	0.111

* Treatments applied between calving and first postpartum control.

veterinary assistance for calving recovered completely after one treatment in 7/8 cases. The application of iodine capsules between calving and endometritis diagnosis led to higher clinical cure probability two weeks after the first treatment (odds ratio iodine capsules application: 5.66). These capsules were used in 11.8% (n = 16) of all included endometritis cases by veterinarians as well as by farmers. Half of the iodine capsule applications occurred in cows having suffered from placental retention (n = 8). In total, 62.5% (n = 10) of cases having

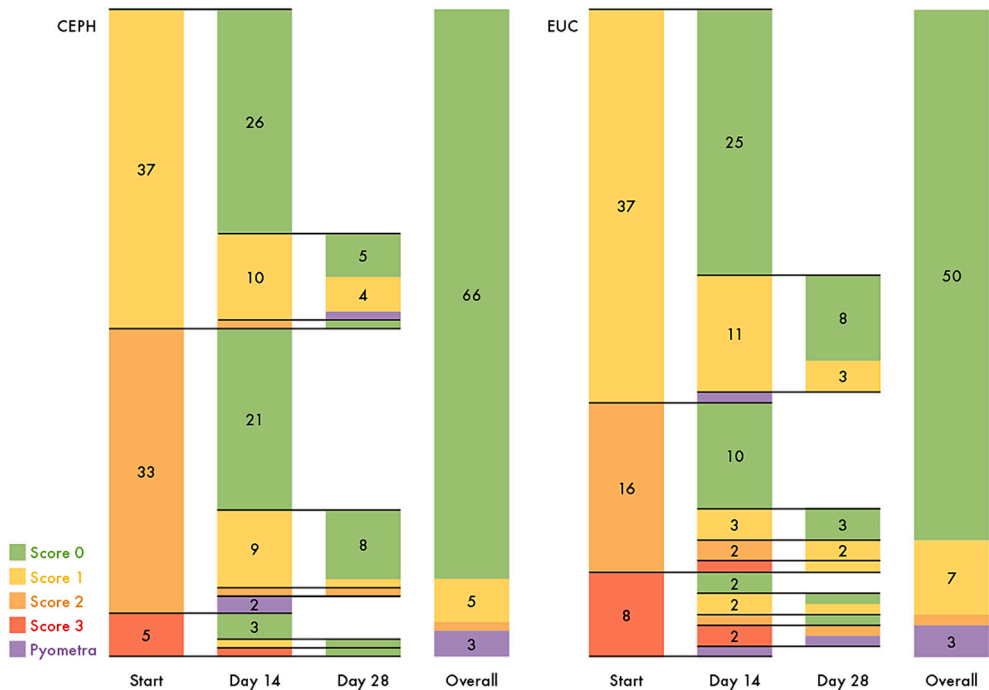


Fig. 2. Disease course from diagnosis to the end of the observation period 14 ± 2 days after the maximum of 2 consecutive treatments. Score 0–3 and pyometra (endometritis classification referring to Sheldon et al., 2006). Column sections that are not labelled with numbers each represent one single case.

received these iodine capsules between calving and endometritis diagnosis showed a complete clinical recovery after one treatment.

3.2.2. Overall clinical cure rate and clinical cure probability

In total, 15% of all treated endometritis cases stayed uncured after the application of a maximum of two consecutive treatments leading to an overall clinical cure rate of 85%. This percentage is slightly lower in the group EUC (82%) than in the group CEPH (88%; at maximum 28 (± 4) days after first treatment). Considering the actual sample size and the detected difference between treatments, the power to show non-inferiority with a maximum difference of 20% was 73%.

A pyometra was diagnosed in two cases 2 weeks after the second treatment (EUC: 1 case, CEPH: 1 case). Numerically, cows with an initial lower endometritis severity (grade 1 and 2) showed higher clinical cure rates than such with an endometritis grade 3 (Fig. 2).

No statistically significant difference could be observed regarding overall cure rate between both treatment groups (Table 3; $\chi^2(1) = 0.009$; $p = 0.923$). Endometritis severity did not affect the overall cure probability ($p = 0.665$). Moreover, none of the covariables tested had a statistically significant effect on overall cure rate.

4. Discussion

The objective of this study was to compare the short-term effect of two registered veterinary medicinal products for intrauterine use, for the treatment of clinical endometritis in dairy cows. It was the first study that compares under field conditions both preparations, the antimicrobial (CEPH) and the herbal product (EUC) and it was the first direct comparison of EUC with an antimicrobial substance. CEPH is the only antibiotic product commercially available in Switzerland for intrauterine use in cases of endometritis. We hypothesized that the intrauterine herbal application is non-inferior to the antimicrobial product regarding the course of clinical signs during a maximum of four weeks after one initial and maximal two consecutive treatments.

The study had a multicentre design and reduced therewith the risk of the fact that treatment effectiveness in farm practice depends on

multiple extrinsic factors (Tamminen et al., 2018).

The recorded clinical endometritis prevalence in our study (21.4%) fits well into the range of prevalence of 17–54% observed in other studies based on the same disease definition and diagnostic time period postpartum (Schlegl et al., 2020; Tison et al., 2017; Makki et al., 2017; Barański et al., 2022).

Most studies on intrauterine treatment of clinical endometritis with cephalixin (Runciman et al., 2009; Denis-Robichaud and Dubuc, 2015; Schlegl et al., 2020) or the herbal product (Schnellbach, 1991; Tischer, 1998; Handler et al., 2005) focused on reproductive parameters and not on clinical resolution of endometritis. Nevertheless, detection of vaginal pus represents the main reason of postponing insemination due to repetitive treatments (Schlegl et al., 2020). In addition Lambert et al. (2014) show that cows with clear vaginal discharge had statistically significant higher non-return rates at 90 days than cows with purulent vaginal discharge. However, further studies with long term observations are needed to test the non-inferiority of EUC compared to CEPH with regard to reproductive parameters.

There are only a few studies, which determined clinical cure rates of clinical endometritis after a maximum of two consecutive intrauterine applications (Barański et al., 2022). However, this therapeutic regimen fits well to the usual biweekly rhythm of farm visits of herd management programs.

A second treatment led to clinical cure in 59.2% ($n = 29$) of the 49 cases diagnosed as uncured after one treatment (Fig. 2). This percentage is near to that obtained by Barański et al. (2022) where a second treatment allowed reaching a clinical cure in 58.8% of uncured cases after one treatment. However, due to a missing untreated control group in our study, it is impossible to differentiate between the treatment effect and the effect of time. Nevertheless, the results of Barański et al. (2022) suggest, that the time factor itself plays an important role.

The exclusion of 33 treated endometritis cases (10 CEPH and 23 EUC; Fig. 1) prior to the statistical analysis were mainly caused by early insemination of many cases (2 CEPH, 11 EUC), lead to a disbalance between the number of cases in each treatment group and reduced statistical power of the outcome. Detected oestrus signs having convinced farmers to inseminate their cows suggested a resumption of cyclicity, which is known to be correlated with uterine health (Gilbert, 2019; Hay et al., 2019; Mohammed et al., 2019).

The trial was conducted without an untreated control group as in cases of clinical endometritis a treatment is expected on commercial farms and expected clinical cure is connected with less economic losses. The absence of a control group is a weak point of our study but non-inferiority testing is a common study design in cases of available registered medicinal products (Walker and Nowacki, 2011; Hahn, 2012). Non-inferiority of EUC compared to CEPH has been shown within the predefined framing conditions, however with a slightly lower power (73% instead of expected 80% for overall clinical cure rate) due to slightly reduced analysable case numbers. Nevertheless, compared to other studies providing information about clinical cure after endometritis treatment with cephalixin (Mari et al., 2012; Madoz et al., 2017; Makki et al., 2017) or the herbal product (Schnellbach, 1991; Handler et al., 2005) our study size fits well in the range of number of endometritis cases per treatment group (about 25–240) and the number of involved farms (1–27; Appendix 1 and 2).

It is to note that endometrial cytology (samples obtained by uterine flushing or through cytological brushes) could have been an interesting additional diagnostic method to confirm the resolution of endometritis clinical cure already assessed through the evaluation of vaginal discharge (McDougall et al., 2020). However, for veterinary practitioner, cytological examinations are rather reserved for cows with suspicion of a subclinical endometritis.

In former studies, clinical cure (absence of purulent vaginal discharge) was slightly better in cephalixin compared to untreated or placebo control (LeBlanc et al., 2002b; Mari et al., 2012; Makki et al., 2017; Tison et al., 2017) but only significant in one study (Mari et al.,

Table 3

Odds ratio (OR), 95% confidence intervals (CI) and p-values for the overall clinical cure rate regarding treatment, pre-treatment endometritis severity and anamnestic parameters.

	OR	Lower 95% CI	Upper 95% CI	p- value
Intercept	8.718	1.713	64.037	
Treatment CEPH	1.051	0.377	2.897	0.923
Endometritis severity score 2	0.893	0.289	2.935	
Endometritis severity score 3	0.500	0.114	2.429	0.665
Retained fetal membranes without fever	0.531	0.100	3.155	
Retained fetal membranes with fever	0.468	0.015	8.699	0.758
Lactation number	0.990	0.772	1.318	0.938
Calving season spring	0.462	0.073	2.565	
Calving season summer	0.748	0.118	4.088	0.796
Calving season fall	0.564	0.103	2.242	
Calving assistance (assistance by farmer)	1.675	0.148	231.925	
Calving assistance (veterinary assistance)	4.769	0.506	661.678	0.428
Twins	1.327	0.275	9.769	0.741
Intrauterine application iodine capsules*	2.182	0.447	16.581	0.358
Antibiotic therapy for uterine indication*	1.189	0.114	25.721	0.896
Systemic antibiotic therapy for another indication than uterus*	0.804	0.108	13.296	0.851
Intramuscular application of long-acting oxytocin*	0.382	0.086	1.674	0.196

* Treatments applied between calving and first postpartum control.

2012) and not in others (LeBlanc et al., 2002b; Makki et al., 2017; Tison et al., 2017). The herbal product was only compared with non-antibiotic intrauterine standard treatments and mostly investigating the effect on reproductive parameters (Heuwiesser et al., 2000; Handler et al., 2005; Kaya, 2008).

Clinical cure rates obtained in our study 14 (± 2) days after the antibiotic treatment are with 67% within the range of former studies (73% (Dohmen et al., 1995), 60% (Madoz et al., 2017), 62.2% (Makki et al., 2017), 75.9% (Tison et al., 2017), 76.4% (Barański et al., 2022)). Clinical cure rate 14 (± 2) days after herbal treatment was 61% compared to 58% (30 of 52 cows) determined 10 days after one treatment in the study by Handler et al. (2005). Overall clinical cure rates obtained in our study 28 (± 4) days after at maximum two consecutive antibiotic treatments are with 88% comparable to Barański et al. (90%; Barański et al., 2022). No data are available to compare the overall clinical cure rate of 82% after two consecutive EUC treatments.

Based on the collected parameters of our study the reason for the non-inferiority of EUC compared to CEPH remains speculative. However, while the mode of action of CEPH is mainly antibacterial (PubChem; Shahbaz, 2017), the multi-component-composition of the four plant extracts of EUC is well known for their multiple activities, in particular anti-inflammatory, antioxidant, immunomodulatory and antimicrobial properties (ESCOP, 2019; ESCOP, 2013; Sharma et al., 2021; Bina and Rahimi, 2017).

Brodzki et al. (2020) demonstrated the negative effect of intrauterine cephalosporin on local bovine uterine immune defences through the inhibition of the neutrophils and macrophages phagocytic activity. Additionally, a detrimental action of antibiotics on uterine microbiome was proved by Molina et al. (2020). No such studies are available with regard to EUC.

Even if the uterine self-healing ability is reported in many studies (LeBlanc et al., 2002a; Giuliadori et al., 2017; Gautam et al., 2010; Plöntzke et al., 2011) the use of local antimicrobial therapy of clinical endometritis is still common in daily veterinary practice (Hehenberger et al., 2015; Goshen and Shpigel, 2006; Runciman et al., 2009; LeBlanc et al., 2002b). The scientifically proven non-inferiority of clinical cure achieved after intrauterine application of an herbal veterinary medicinal product might therefore represent a benefit in the daily veterinary practice with regard to the targeted reduction of antibiotic use in dairy cattle. The herbal veterinary medicinal product contributes at least to satisfy the farmers and veterinarians request of treating cows suffering from purulent vaginal discharge instead of letting them untreated.

Statistical analysis of our data revealed the significant impact of three fixed effects 14 days after treatment: retained fetal membranes as a risk factor for clinical cure, while veterinary assistance due to abnormal calving and intrauterine application of iodine capsules between calving and treatment of clinical endometritis promote the clinical healing (Supplement, Tables 2 and 3). Even if these findings are not based on a high number of cases they are well in line with former studies (Knutti et al., 2000; LeBlanc, 2008; Dubuc et al., 2010; Mock et al., 2015; Geiser et al., 1995; Mido et al., 2016; Yoshida et al., 2020). It may underline that our sample represents overall a usual study population. Breed distribution is not fully balanced in our study population as the Holstein breed is slightly overrepresented in EUC and Brown Swiss in CEPH. However, pure milk breeds (Holstein, Brown Swiss, Jersey – together 80% in CEPH vs 82% in EUC) and dual purpose breeds (Montbéliarde, Swiss Fleckvieh – together 20% in CEPH and 18% in EUC) are well-balanced between the treatment groups.

Interestingly, endometritis severity score did not statistically influence the overall clinical cure rates. This might be due to the small number of animals with different vaginal discharge scores. In more epidemiological directed studies with large numbers of animals, it was found that odds for clinical cure decreased significantly with higher vaginal score (Madoz et al., 2017; Gautam et al., 2010).

5. Conclusion

In conclusion, the clinical cure of dairy cows' endometritis after the intrauterine application of an herbal product based on *Calendula officinalis* L., *Melissa officinalis* L., *Origanum majorana* L. and *Eucalyptus globulus* Labill. was non-inferior to the intrauterine application of the standard antibiotic cephalosporin under field conditions. Regardless of the controversial discussion about the fundamental benefit of intrauterine endometritis therapy, our results may at least encourage to reduce the use of antimicrobials in the daily veterinary routine treatment of bovine endometritis. Finally, this study may stimulate more clinical research projects on phytotherapy in the context of dairy cow's health management.

CRedit authorship contribution statement

Valérie Menoud: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mirjam Holinger:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Sandra Graf-Schiller:** Conceptualization, Funding acquisition, Resources, Writing – review & editing. **Philipp Mayer:** Conceptualization, Funding acquisition, Resources, Writing – review & editing. **Luc Gerber:** Writing – review & editing, Software, Investigation, Data curation, Conceptualization. **Michael Walkenhorst:** Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Gaby Hirsbrunner:** Conceptualization, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rvsc.2024.105250>.

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